

ENGINEERING DEVELOPMENT FOR SODIUM SYSTEMS

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1. Introduction

The Prototype Fast Reactor (PFR)⁽¹⁾ will be cooled with liquid sodium, and to support the design of the primary and secondary coolant circuits, the UKAEA have mounted an extensive programme of engineering development work. Starting from the experience gained from the Dounreay Fast Reactor, this programme is designed to provide the basic technology necessary for the design and operation of sodium circuits at temperatures up to 650°C and fluid velocities up to 5 m/sec.

In addition to this essential background of sodium technology experience, the engineering development programme aims to provide information for the design of individual components where there are insufficient data at present available for this to be done with complete confidence. To allow the designer freedom to modify details as the design evolves, the approach has been to obtain broadly based information on the specific aspects where data are lacking; complete components are tested only where this is found to be the most effective means for deriving the information or where, as in the development of the large mechanical pump, operational experience is particularly desirable.

Thus, for mechanisms which have to work either submerged in the sodium pool or suspended in the cover gas above it, experimental information is being obtained on the friction and wear characteristics of materials in the sodium environment and on the performance of a range of bearing designs. For the steam generators, experiments are being done to provide the designer with information on the possible consequences of the reaction between steam and water.

These are examples of development problems which result from the chemical properties of sodium and thus require experiments in sodium. The hydraulic performance of the sodium circuits depends on the physical

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properties of sodium, and the relevant conditions may be modelled adequately in more easily handled fluids, such as air or water. Examples of the design problems handled in this way are the evaluation of the hot spot factors of subassemblies; and flow, vibration, and gas entrainment in the sodium circuits.

This paper describes these various developments, placing most emphasis on the equipment for sodium experiments and the experience gained from it.

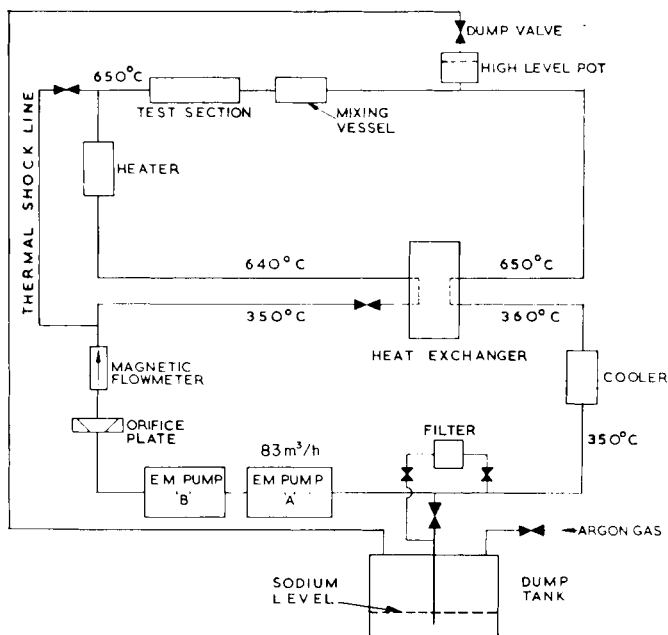


FIG. 1. Fuel element test rig.

2. Sodium Rigs

The basic facilities for this experimental programme are four sodium rigs which have been constructed at the Reactor Engineering Laboratory (REL), Risley. They were designed to produce the sodium environments necessary for the various experiments, and also to provide operational experience on sodium circuits which would be relevant to the PFR. Where possible, construction techniques applicable to the PFR have been applied so as to gain experience in this field also.

Rig Construction

These rigs, which are basically similar, consist of a closed circuit round which the liquid metal is pumped, test sections in which experiments can be

TABLE 1 DETAILS OF UKAEA SODIUM RIGS

Rig title	Location	Commission date	Metal	Pipe diameter (mm)	Flow (m ³ /h)	Metal hold up (m ³)	Max. temp. (°C)	Remarks
Pump test rig	REL	1964	Na	305	1620	14	400	Three test vessels
Fuel element test rig	REL	1963	Na	108	83	3.68	650	
4 in. loop	DERE	1964	70/30	108	54	1.75	600	
1 in. general purpose loop	REL	1963	NaK	30	8.3	0.28	650	
Corrosion loops (nine)	RML	1955-63	Na	18	4.2	0.012	750	
Rubbing pair rig	REL	1963	Na	30	4.2	0.085	650	
Twin zone loop	DERE	1963	Na	18	4.2	0.034	650/700	
$\frac{1}{2}$ in. loop	DERE	1962	70/30	18	4.2	0.014	650	
Components test loop	DERE	1961	NaK	24	8.3	0.11	650	
$\frac{1}{4}$ in. loop	DERE	1962	Na	10	4.2	0.008	650	
Flow calibration rig	DERE	1965	Na	108	400	1.40	600	No pump fitted Eight test vessels
Mechanisms rig	REL	1966	Na	30	2.9	3.68	650	
NOAH	DERE	1963	Na	—	—	0.34	450	

performed, a header tank with a free surface to allow for thermal expansion, a dump tank into which the sodium can be drained, an auxiliary circuit containing a cold trap and impurity monitor, and a system for providing an inert cover gas, usually argon. Figure 1 shows a typical rig.

The main characteristics of the rigs are shown in Table 1, which includes, for comparison, details of liquid metal rigs at the Dounreay Experimental Reactor Establishment (DERE) and the Reactor Materials Laboratory

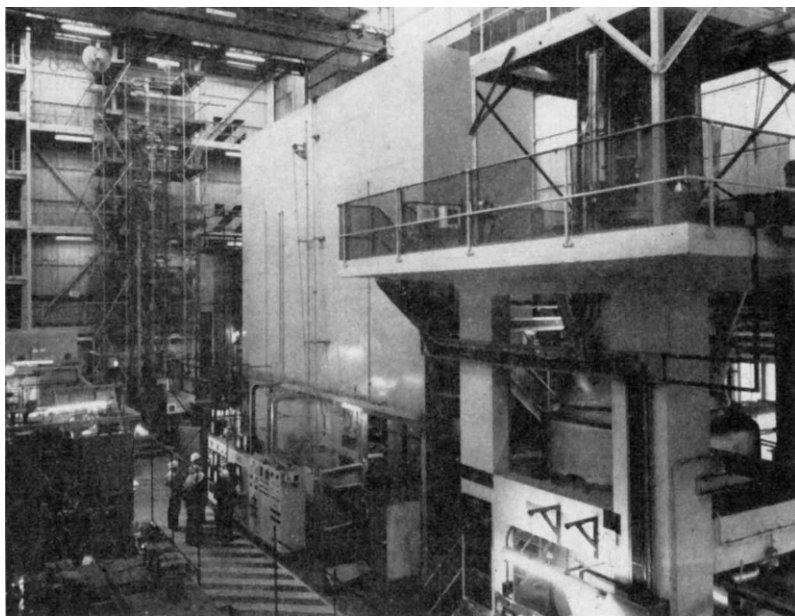


FIG. 2. Sodium rig building.

(RML), Culcheth. A general view of the sodium rig building at REL is given in Fig. 2.

The rigs have been designed for maximum flexibility in operation to cope with possible programme changes as the PFR design proceeds, and for maximum reliability, so that lost time caused by rig faults is minimized. The sodium circuits are constructed primarily from 18:8:1 titanium stabilized austenitic stainless steel (similar to AISI type 321) and designed to the appropriate codes⁽²⁾ in so far as they are applicable. Welds which are highly stressed relative to the strength of the material at the operating temperatures are fully radiographed, while a proportion of the remaining welds are radiographed to ensure a continuing high standard of workmanship.

To ensure maximum rig reliability, only proven components are used in the main loops, components under development being restricted to the test

sections where they can be more easily repaired or maintained. To eliminate valve spindle leakage, bellows sealed valves are used exclusively in the sodium circuits, although valves with packed spindle glands may be used in the cover gas circuits. For many years fairly compact valves incorporating edge welded bellows were used, but these proved unsuitable for some applications above 400°C , and a new type of valve has been developed, incorporating

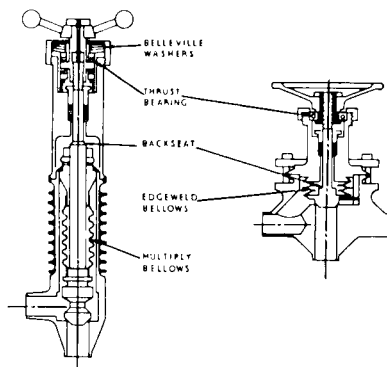


FIG. 3. Sodium valves.

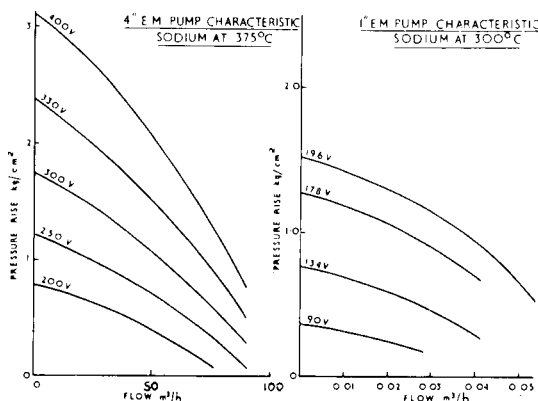


FIG. 4.

convoluted bellows and an extended bonnet and spindle to reduce the thrust bearing temperature. These valves have back seats and a back-up spindle gland to limit the sodium leakage which can occur if the bellows fracture, and can be used up to 650°C in pipes up to 100 mm diameter. Both types of valve are shown in Fig. 3.

Two types of electromagnetic pump are used for circulating sodium round the loops. For small loops, a.c. conduction pumps with the characteristic shown in Fig. 4 are used.⁽³⁾ With class H windings the upper temperature

limit for these pumps is 180°C, but a version having ceramic insulation and a temperature limit of 500°C is being developed. For larger rigs, flat linear induction pumps similar to those in DFR⁽⁴⁾ are used, with improved high temperature insulation to enable them to run at sodium temperatures up to 375°C. Their characteristics are shown in Fig. 4.

Leak Testing

The high standard of rig design and manufacture described above ensures that major pipe or vessel failures are extremely unlikely. Small sodium leaks, caused, for example, by a valve bellows failure or a pinhole in a weld, do occur, and the rig operators are protected from these by enclosing the whole rig in a sheet metal cubicle whose floor is composed of individual trays each about 1 m square. The rig designs are such that, in the event of a leak occurring, at least 90% of the metal can be quickly drained into the dump tank by opening one valve. The rest of the metal can be drained by opening further valves, and to facilitate drainage all pipe runs have a slope of at least 1:10.

The quantity of sodium escaping is thus limited; it is caught in the trays which prevent it from spreading, and the resulting fire is readily extinguished by smothering the molten metal with dry soda ash.

In low temperature circuits where the liquid metal does not wet the pipes, surface tension will prevent sodium leaking through any hole smaller than about 2×10^{-3} mm diameter. Experience has shown that, even when the sodium is above the wetting temperature, it does not leak through holes of this size, possibly because they are quickly sealed with sodium oxide; consequently, it is REL practice to endeavour to eliminate from sodium circuits all holes greater than this minimum diameter. This is readily achieved on individual components by using vacuum leak detection methods or by pressurizing with a tracer gas and sniffing for leaks, and similar methods are used to detect leaks in each rig weld.

In this way all significant leaks in a rig should be eliminated, but it is highly desirable to make an overall check to ensure that no large leak has been overlooked. On a large circuit vacuum methods are tedious, and the pressure decay method is subject to appreciable errors from ambient temperature changes. A novel method of overall leak detection using an isotope abundance technique has been developed;⁽⁵⁾ it is a significant improvement on these existing techniques and is applicable both to rigs and reactor circuits.

The circuit to be leak tested is first filled with argon to the working pressure, and a known amount of helium-4 is then introduced into the system. A typical concentration is 50 ppm, but the actual value depends on the sensitivity of the mass spectrometer used for the quantitative measurements. After an appropriate time interval a known amount of helium-3 is introduced

into the system. The abundance ratio of the two isotopes is then measured using a mass spectrometer, and the total gas outleakage in the interval between the two injections of helium is evaluated.

Making the basic assumption that the leakage of helium-4 represents the total loss from the system, i.e. that there has been perfect mixing and that all leaks are homogeneous, the sensitivity of the technique is limited by the accuracy with which known amounts of the helium isotopes can be dispensed. With present techniques an overall leakage of 1% of the total circuit volume can be measured; consequently, if the test is extended over 5 days, a leak rate of 0.2% per day can be detected. If the leak were concentrated at one place in a circuit having a volume of 20 m³ this rate would correspond with a hole 0.1 mm in diameter. Thus, while this technique is a significant advance over those previously used for measuring an overall leak rate, it cannot compare in sensitivity with leak testing of individual components.

Sodium Purity Measurement and Control

The REL rigs have been designed to provide information on the amount of impurity which is likely to be present in the PFR circuits. Particular attention is being paid to oxygen, hydrogen, calcium, and carbon, which can enter by the following routes:

- (a) impurities present in the sodium as received;
- (b) dirt present in the components after cleaning;
- (c) dirt entering the circuit during construction;
- (d) impurities in the inert cover gas; typical impurities in argon are 10 ppm of oxygen and water vapour;
- (e) impurities entering the rig during maintenance.

The amount of impurity likely to enter the circuit from the surfaces of the components has been assessed by measuring the impurity content remaining in various structural materials after cleaning. Samples of the materials were fused under vacuum and the impurities thus released were determined. Typical values are:

	18:8:1 stainless steel	2½%Cr-1%Mo steel
Oxygen	2.8×10^{-5} g/cm ²	62×10^{-5} g/cm ²
Hydrogen	1.3×10^{-5} g/cm ²	0.5×10^{-5} g/cm ²

Information about the amounts of impurity likely to enter the PFR circuit by the routes mentioned above can be obtained by monitoring the impurity content of the rigs during start-up and subsequent operation. The most useful

rigs in this respect are the mechanical pump rig, whose dump tanks are made from $2\frac{1}{4}\%$ Cr-1%Mo steel and whose parameters (ratio of surface area to volume, etc.) are similar to those of the PFR, and the fuel element thermal test rig which attains a sodium temperature of 650°C .

The total impurity level in the rigs is determined conventionally using

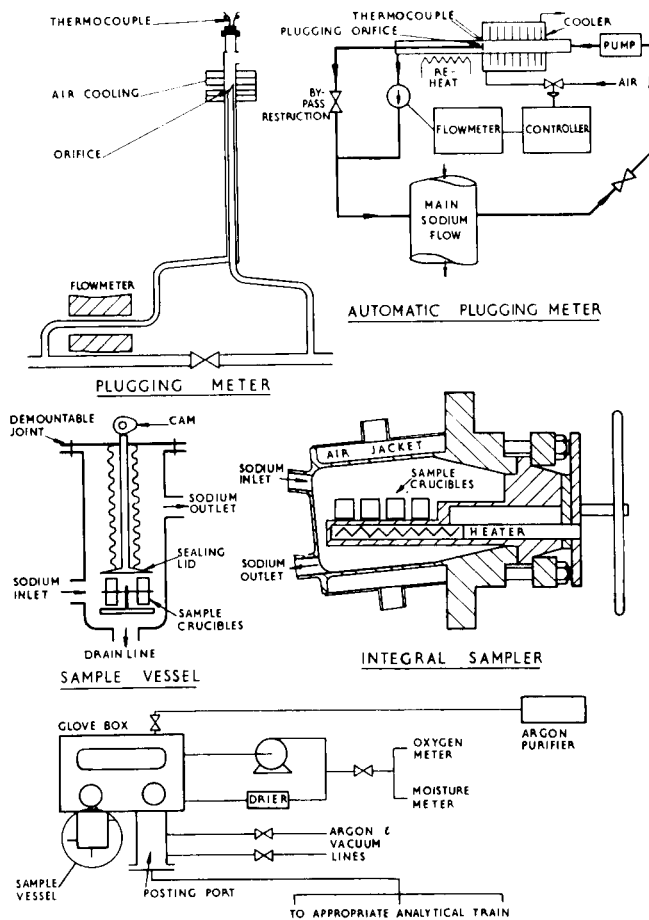


FIG. 5.

manually operated plugging meters.⁽⁶⁾ An automatic plugging meter which gives a continuous record (Fig. 5) has also been developed and is being used on some rigs.⁽⁷⁾ When a combination of impurities is present, however, the information provided by either of these instruments is difficult to interpret; therefore, to obtain a better understanding of the behaviour of impurities, the technology of sampling has been developed. In the method now used, a

sample of sodium is taken by lowering some small, lidded pots below the sodium surface, and filling them with sodium (Fig. 5). The samples are then lifted into a glove box in which they are transferred to a distillation pot, the sodium is distilled away, and the residue is chemically analysed. To avoid contamination of the sample by oxygen and water vapour, the argon atmosphere in the glove box is carefully purified by circulating it through molecular sieves, and provided that precautions are taken to obtain a representative sample, an accuracy of better than 10 ppm can be achieved in estimating the concentration of oxygen,⁽⁸⁾ hydrogen,⁽⁹⁾ carbon,⁽¹⁰⁾ and calcium.

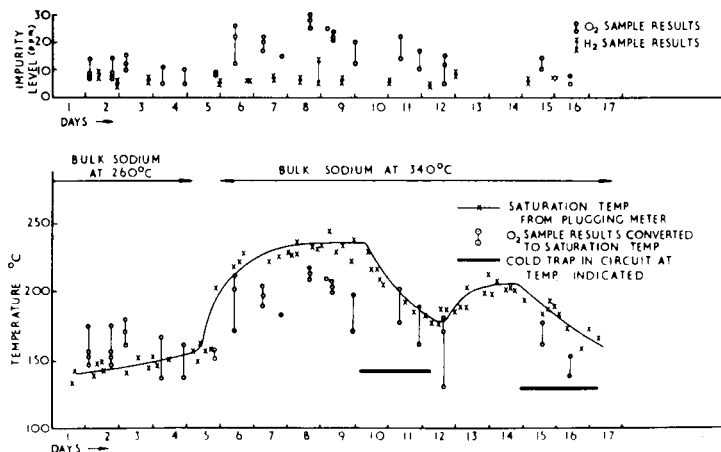


FIG. 6. Typical clean up run.

In this method the sodium samples are removed from the rig for distillation because this enables existing equipment to be used. An improved sampler (shown in Fig. 5) in which distillation takes place in the rig itself has been developed and is being introduced on to the rigs. By using these techniques the behaviour of impurities in the rigs can be followed in detail, and Fig. 6 shows some information obtained from a typical cleanup run on the mechanical pump rig, which contains 14 m³ of sodium and has 150 m² of pipework. In an effort to limit contamination during construction to that which is likely to obtain on the reactor, the rigs are assembled in the sodium rig building, where reasonable cleanliness is maintained. The various rig components are fabricated elsewhere, degreased, pickled, washed and delivered in polythene bags. The pipes are closed by removable blanks until welded in position.

When a sodium circuit is to be filled, the rig pipes are trace heated to about 200°C to provide an adequate margin above the freezing point of sodium, and the sodium in the dump tank is heated to 140°C, this being a compromise between making the sodium reasonably fluid and avoiding taking

too much impurity into solution. The sodium is then pushed up by gas pressure into the rig through a filter, which prevents any gross amount of solid impurity entering the circuit. Once in the rig the sodium begins to dissolve additional impurities from the rig walls, and the pumps are started to prevent blockages forming and to allow the impurity level to be measured.

The impurities are controlled in the conventional manner by cold traps,^(11, 12) and by measuring the variation of impurity content with time at various rig and cold-trap temperatures it is possible to estimate both the rate of increase of impurity level and the total impurity taken up under various conditions. Figure 6 illustrates the progress of a typical cleanup run, and shows that reasonable consistency is achieved between the sample analyses and the measured plugging temperature. Results from such runs indicate that when cleaning a newly constructed stainless steel rig about 10^{-4} g of oxygen per cm^2 of wetted pipe area were removed by the sodium. A comparison of this figure with the amount (2.8×10^{-5} g/ cm^2) released during vacuum fusion shows that a significant amount of oxide contamination had entered the rig during construction and filling.

3. Component Development

Sodium Pump

The sodium rigs which have been described are being used in the development of a number of the major PFR components. Basic design information for the PFR primary and secondary pumps⁽¹⁾ is being provided by a prototype mechanical pump, designed by the Authority and being tested in the mechanical pump rig.⁽¹³⁾ The general arrangement of the pump is shown in Fig. 7. Its characteristics are:

Delivery	1620 m^3/h
Head	4.2 kg/cm^2
Speed	980/100 rev/min
Power	475 h.p.
Overall length	7 m
Impeller diameter	740 mm
Specific speed (British units)	1600
(m^3/sec , m, rev/min)	35

Pump Design

The vertical pump shaft, which is 5 m long, is supported in two bearings. The top bearing, which is oil lubricated, consists of two opposed taper roller bearings, and rubbing face seals having Stellite and lead bronze mating faces provide oil seals both to atmosphere and to the argon cover gas. The seals and bearings are mounted on a common sleeve, an arrangement which allows them to be handled as a complete unit and simplifies maintenance.

The shaft bottom bearing, which is below the sodium surface, is sodium lubricated, and a hydrostatic or jetted bearing fed from the pump delivery has been chosen for this application, primarily because it permits a larger radial clearance than the alternative hydrodynamic type of bearing. The theory of hydrostatic bearings under the operating conditions of interest here, which include both laminar and turbulent flow régimes in the bearing

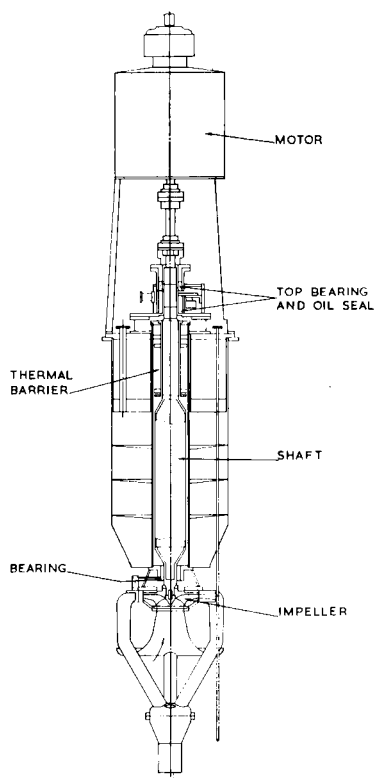


FIG. 7. Sodium pump.

clearance, has been developed and verified by experiments using water and paraffin as the working fluid.⁽¹⁴⁾ A bearing 200 mm diameter $\times$ 200 mm long, having a radial clearance of 0.28 mm and sixteen jets of 4.3 mm diameter, was chosen (Fig. 8); it is made from stainless steel with spray fused Stellite-6 deposited both on the bore of the bearing and on the shaft. When operating at maximum supply pressure, the sodium flow through the bearing is 16.2 m³/h.

To limit the unbalanced radial fluid forces acting on the centrifugal type pump impeller, the sodium is discharged at four points arranged symmetrically around the diffuser casing. These four discharge ducts turn through a right angle, join at a common header and finally discharge vertically downwards. The pump is powered by a 475 h.p. commutator-type, a.c. motor

controlled through an induction regulator to provide infinitely variable speed over the range 10% to full speed.

The mechanical pump test rig consists of a 2 m diameter vessel in which the pump operates submerged in the sodium, and a loop of 300 mm diameter pipe. The pump tank simulates the expected reactor conditions in that it provides a similar method of support for the pump, a free surface of sodium, and a vertical discharge. The supporting structure was designed to have a

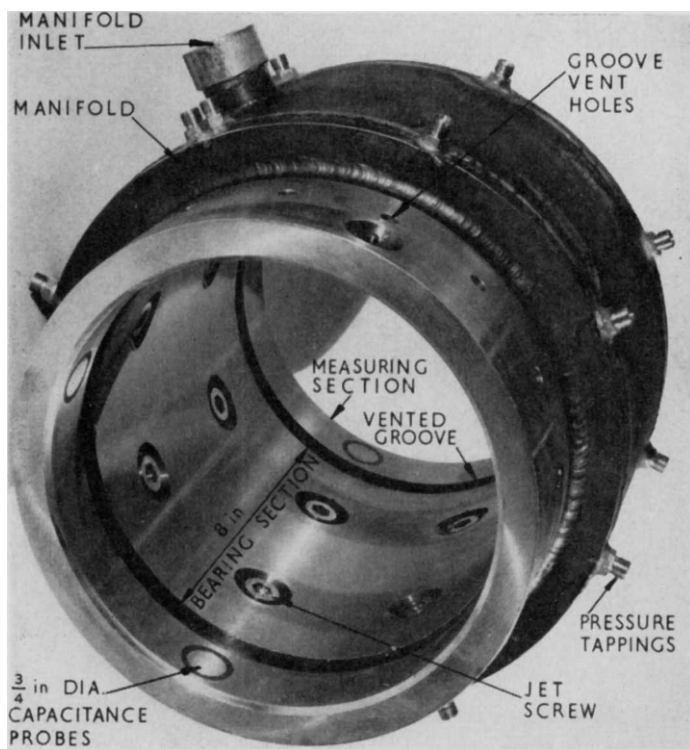


FIG. 8. Pump bearing.

natural frequency above the pump running speed, and baffling is provided inside the tank to simulate reactor inlet conditions. The loop of 300 mm pipe, fabricated in stainless steel, contains a flow control valve, a Venturi flowmeter and an electromagnetic flowmeter. A second pipe of mild steel surrounds the 300 mm pipe and the pump tank, to form an annulus through which air can be circulated. This air can be heated by finned electrical resistance heaters to raise the loop temperature to 200°C before filling with sodium, or, alternatively, cold air can be pumped through the annulus to remove the heat generated by the pump. The mild steel jacket also serves as secondary

containment, providing protection for the operators in the event of sodium leakage.

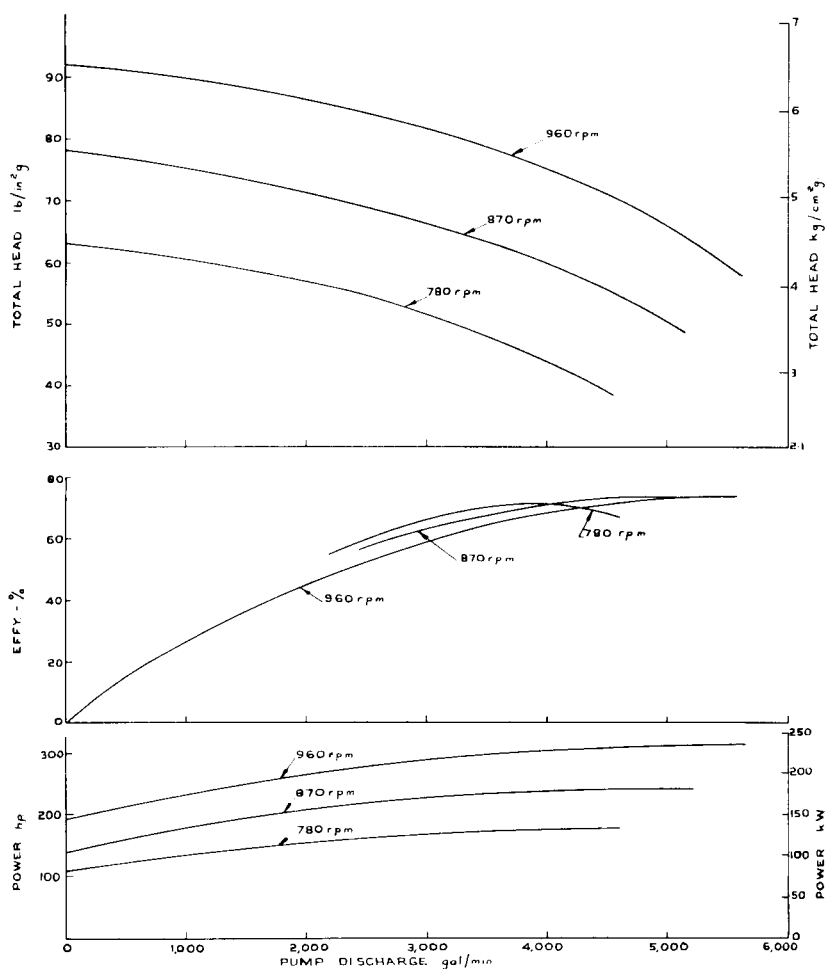


FIG. 9. Mechanical pump performance curves.

Pump Performance

The pump was started for the first time in November 1964 and removed for examination in September 1965. During this period it ran faultlessly for over 3000 hours at various speeds between half and full speed, and was stopped and started over 250 times. The oil leakage past the shaft seals decreased from 1 cm³/h at the beginning of the tests to 0.2 cm³/h during the

final run. The measured hydraulic performance of the pump is shown in Fig. 9.

During the tests a significant amount of sodium mist was seen above the sodium pool, and some solid sodium was deposited in the sight tubes which penetrate into the main pump vessel, but when the pump was removed only minimal quantities of sodium were found on the pump components. No significant deposition had occurred on those parts which were above the sodium level, and those below the sodium had drained almost completely.

While the pump was being partially dismantled before steam cleaning, some difficulty was experienced in removing the bolts and separating those flanges which had been immersed in the sodium. Apparently the liquid metal had penetrated into the fine clearances between the mating parts and was causing some form of adhesion even when heated above its melting point, although when separated the mating surfaces showed no sign of pick-up or diffusion welding. This information will influence the detailed design of the PFR pump: in particular, jacking bolts will be fitted to facilitate the separation of flanges.

After steam cleaning the pump was thoroughly examined. The sodium lubricated bottom bearing was found to be in perfect condition, with no sign of wear on either the shaft or the sleeve. In the oil lubricated top bearing the oil seals were in excellent condition, but the unloaded taper roller bearing showed a small amount of cage wear, possibly caused by incorrect clearance in the assembly. There was minor cavitation damage to the pump impeller and diffuser, possibly caused by running the pump away from the design point during measurements of the pump hydraulic performance.

4. Sodium Lubricated Mechanisms

Basic information to guide the design of the PFR fuel handling mechanisms, and other mechanisms which will have to operate submerged in the sodium pool, is being obtained by an extensive study of the friction and wear characteristics of suitable bearing materials. A crossed cylinders apparatus⁽¹⁵⁾ is used in which one specimen, in the form of an 8 mm square section pin, is axially loaded and held stationary against a 50 mm diameter sleeve specimen which is rotated at constant speed. A cylindrical curvature of 4 mm radius is machined on the rubbing face of the pin specimen. The friction force is continuously indicated. Rubbing speeds are kept low, nominally a few centimetres per second, to minimize frictional heating effects (more especially in gaseous environments, such as argon containing sodium vapour), and the possibility of hydrodynamic lubrication for the liquid metal environment. A number of such crossed cylinder test pots are connected to the rubbing pair rig (listed in Table 1) and supplied with sodium of controlled purity at temperatures up to 500°C. Other test pots are supplied with argon

supersaturated with sodium vapour, the impurity levels in the argon cover gas being controlled so as to vary the amount of sodium oxide deposited in, and around, the sliding region.

Wear of the stationary (pin) specimen is evaluated from the dimensions of the elliptical wear scars, either by direct mensuration or by using an optical interference technique. The wear data are correlated on the basis of the simple theory of adhesive wear,⁽¹⁶⁻¹⁸⁾ and specific wear rates, defined as the volume of material removed per unit distance rubbed per unit applied load ($\text{cm}^3/\text{cm kg}$), are determined. The results of experiments on combinations of Stellites, tool steels and low-alloy steels have been reported;^(15, 19) specific wear rates at 400°C of the order of $10^{-11} \text{ cm}^3/\text{cm kg}$ and coefficients of friction around 0.5 can be obtained using suitable materials.

To enable individual components of the various PFR sodium lubricated mechanisms to be tested under the correct conditions, a new rig (mechanisms rig, Table 1) is being constructed and will supply eight test sections with sodium of controlled purity at 400°C . The first six test sections will be commissioned during the second half of 1966, and two further test sections, capable of accommodating complete control or shut-off rod mechanisms, or major components of the fuel handling machine, will be added in 1967. The test sections can be individually heated, if required, to 650°C .

5. Core Components

Fuel Element Test Rig

The behaviour of the various PFR core components, e.g. core and breeder subassemblies, control and shutoff rods⁽²⁰⁾ in sodium at the maximum fuel can temperature (650°C) flowing at velocities similar to those which will obtain in the PFR core can be investigated in the fuel element thermal test rig. This rig (shown in Fig. 1), which is constructed of 100 mm diameter pipe, was designed to provide $90 \text{ m}^3/\text{h}$ of sodium at 650°C ; to limit the electromagnetic pump circuit temperature to 375°C a design incorporating a heat recuperator and a heater has been used. A by-pass line from the cold circuit to the inlet of the test section enables thermal shocks at a rate of $50^\circ\text{C}/\text{sec}$ to be applied to the test section; a large mixing vessel immediately following the test section protects the heat recuperator from thermal shock.

The heat recuperator, of 10 MW capacity, is of straightforward shell and tube design, the only significant design problem being that of ensuring satisfactory flow distribution through the tube side with the pressure drop available, but the design of the heater posed a problem which has no counterpart in the reactor. It was necessary to supply 400 kW to heat the sodium in the hot loop from the recuperator outlet temperature (640°C) to 650°C , and initially this was done by using seventy stainless steel clad, mineral insulated heaters, each 9 mm diameter $\times$ 4 m long, rated at $20 \text{ W}/\text{cm}$ length ($6 \text{ W}/\text{cm}^2$)

and running at 240 V. These heaters proved unsatisfactory because holes formed in the 0.9 mm thick stainless steel heater sheaths, allowing sodium to flow up the inside of the heaters into the electrical terminal boxes. These failures were probably caused either by inclusions in the heater sheaths which were leached out by the sodium, or by electrical breakdown of the high temperature magnesia insulation, and pointed to the need for more robust containment of the liquid sodium.

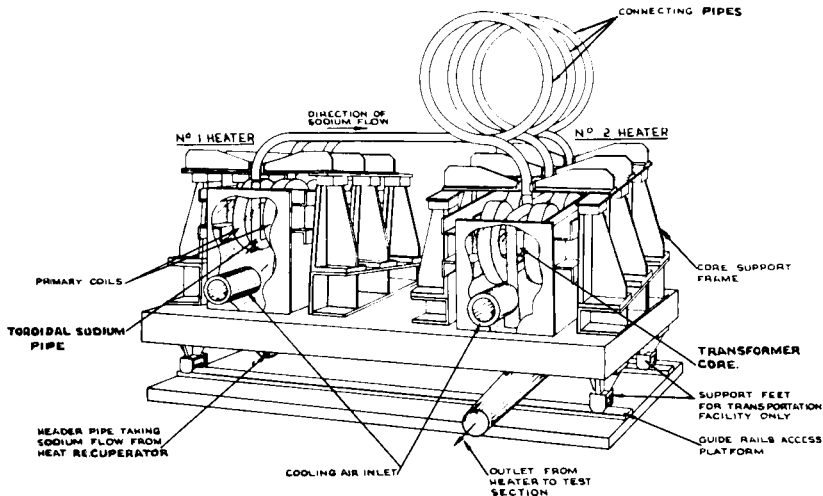


FIG. 10. Induction heater.

This has been achieved in a new design of heater (Fig. 10) in which the heat is generated directly in the liquid metal. The sodium leaving the heat recuperator flows through three 50 mm diameter toroidal pipes which form the single shorted turn secondary windings of a three-phase transformer. These pipes are at 650°C and are thermally insulated from the primary windings, which are conventionally wound using class H insulation and air cooled. Because the electrical resistivity of the liquid sodium is considerably less than that of the stainless steel pipe containing it, most of the secondary current flows through the sodium, generating 200 kW of heat at a power factor of 0.83. Two heaters are placed in series to develop the 400 kW required, the power of one of them being infinitely variable through a voltage regulator in the primary supply circuit. This design of heater has the great merit that the walls of the toroidal pipes can be at least as thick as those of the connecting pipework and all the pipe welds can be fully radiographed. Thus the standard of sodium containment is at least as good as it is in the remainder of the rig, and the risk of sodium leakage has been substantially eliminated.

A PFR core subassembly is being tested in the rig, and over 1000 hours of satisfactory operation have been achieved.

Core Flow and Temperature Distributions

Although the correct sodium environment is necessary to investigate some aspects of the mechanical behaviour of the PFR core components, the

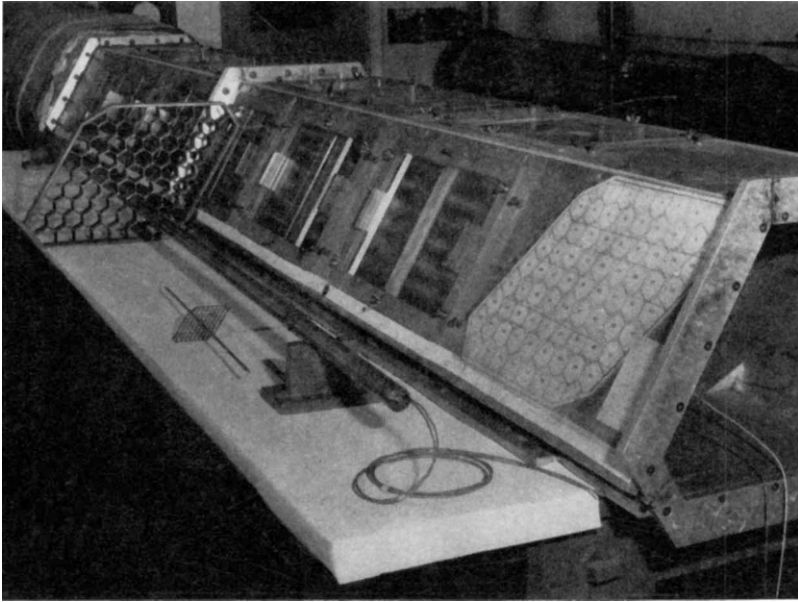


FIG. 11. Model of fuel subassembly.

flow and temperature distributions in the fuel subassembly can be investigated using more convenient fluids. The detailed flow pattern and pressure losses within the subassembly are being examined using a $5\frac{1}{2}$ times full scale model (Fig. 11) of a section of the fuel bundle built to the same absolute geometrical accuracy as the prototype. Air at reactor Reynolds number is used as the working fluid, and, as shown in Fig. 12, flow distributions appropriate to perfect geometry and to practical geometries resulting from engineering tolerances have been measured using miniature Pitot tubes.

From these flow distributions an initial temperature distribution can be calculated; this is smoothed out both by molecular conduction, whose effect is readily assessed theoretically, and fluid mixing, whose effect has to be determined experimentally. The magnitude of the fluid mixing is assessed, also on the $5\frac{1}{2}$ times full scale model, by injecting nitrous oxide as a tracer

gas into various points in the coolant channels and measuring its concentration downstream.⁽²⁰⁾

To determine the effect of this fluid mixing on the temperature distribution, it is necessary to assess how much heat is transported by the fluid. Because the molecular conductivity of the sodium is large, theory predicts that, over a significant range of Reynolds number, the heat transported by eddies will

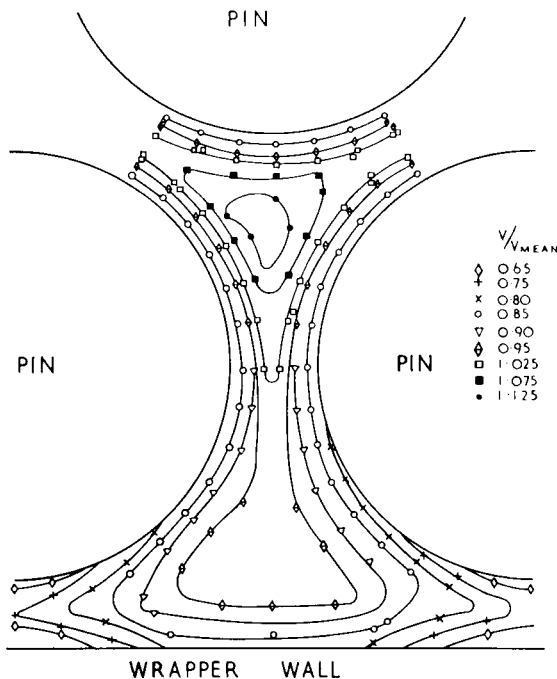


FIG. 12. Velocity profiles in fuel subassembly.

be significantly less than that transported in a non-conducting fluid.⁽²¹⁻²³⁾ It is therefore necessary to determine the ratio of the eddy diffusivities of mass in air, $\epsilon_{M(\text{air})}$, to heat in sodium, $\epsilon_{H(\text{Na})}$, and this is being done by measuring these two quantities separately in identical configurations.

The quantity $\epsilon_{M(\text{air})}$ is measured by injecting a stream of nitrous oxide into the axis of an 80 mm diameter tube carrying a stream of air, and measuring the concentration profiles downstream of the injection point. The quantity $\epsilon_{H(\text{N})}$ is measured in a separate test section, attached to the fuel element test rig, in which a stream of sodium at 600°C is injected into the centre of an 80 mm diameter pipe carrying sodium at 200°C, the stream velocities being suitably matched. By traversing 0.5 mm diameter insulated junction stainless steel sheathed thermocouples across the 80 mm pipe, the temperature profiles downstream of the injection point are measured (Fig. 13), and

from these and an accurate measurement (to about $\pm 5\%$) of the sodium flows, $\epsilon_{H(\text{Na})}$ is calculated. Preliminary data from the experiments indicate that $\epsilon_H/\epsilon_M = 0.6 \pm 0.1$ over the range of Reynolds numbers of $2 \sim 6 \times 10^4$.

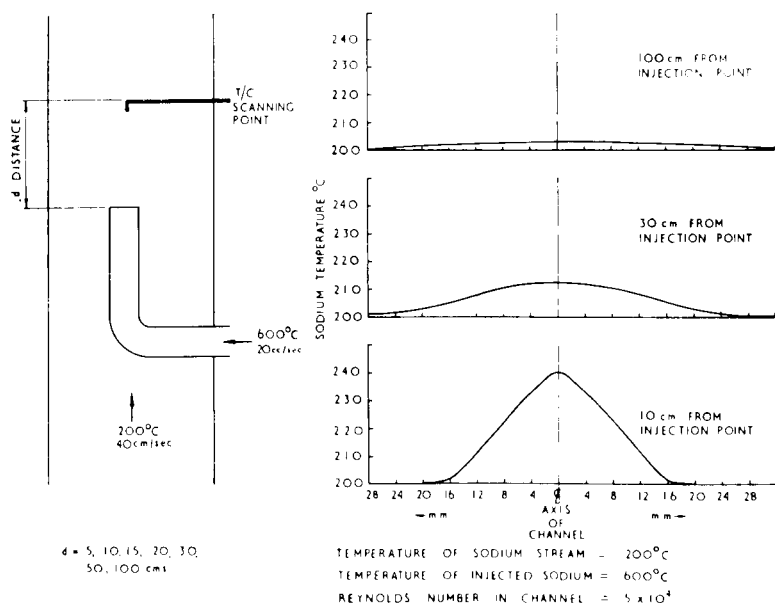


FIG. 13. Eddy diffusivity experiment.

Reactor Vessel Flow Patterns

The flow patterns in the reactor vessel above the core are being investigated using small scale models with water as the working fluid. The free sodium surface in this part of the coolant circuit introduces problems of wave and vortex formation in the surface and the possible entrainment of gas into the coolant stream. Dimensional analysis shows that, besides Reynolds number, both Froude and Weber numbers require simulation, and this is impracticable using water. Consequently a compromise is adopted, whereby gross surface effects are examined in small models in which only Froude number is simulated, while for specific gas entrainment investigations models built to a 5/8 scale are used, and both Froude and Weber numbers are simulated in these.

Using models to 1/17 and 1/5 scales attached to a water loop which will supply 1800 m³/h at 7 kg/cm² head, it has been shown that the unhindered jet of fluid issuing from the core raises a 500 mm high hump at the sodium surface. This would give rise to unacceptable gas entrainment. These models are being used to develop arrangements of baffles which will be fitted to the reactor vessel to eliminate this effect.

6. Steam Generators

When designing the PFR steam generators it is necessary to decide on a shell design pressure which is practicable, economic, and adequate to withstand the effects of the sodium/water reaction which would result from the simultaneous failure of several water tubes. Information on which to base this decision is being obtained from experiments at DERE in the NOAH rig (see Table 1), which is a 1/6 scale model of an early design of PFR steam generator. Related information is being obtained by workers elsewhere.⁽²⁴⁾

A tube terminated by a bursting disk is located within a simulated tube bundle inside the rig vessel, which is filled with sodium at 300°C. The sodium/water reaction is initiated by opening a valve connecting the central tube to a supply of water at 140 kg/cm²; this ruptures the bursting disk, allowing water to stream into the sodium for as long as the valve remains open. The rig parameters are:

Vessel diameter	600 mm	Water pressure	140 kg/cm ²
Vessel height	2500 mm	Water temperature	300°C
Mass of sodium	300 kg	Mass of water injected	~20 kg
Sodium temperature	300°C		

Ten experiments to investigate sodium/water reactions have so far been completed:

Experiment no.	Diameter of water injection hole	Type of test section
1	3.8 mm	1/6 scale tubes
2, 3	15 mm	Single full size tubes
4 to 8, 10	15 mm	Full size tube inside bundle
9	1.6 and 3.2 mm	Full size tube inside bundle

Experiment No. 1 revealed some defects in the apparatus; the thermocouples and the pressure transducers, which had been placed close to the point of tube failure, proved unable to withstand the reaction conditions. In later experiments more robust thermocouples were used, and the pressure transducers were replaced by strain gauges fitted to the outside of the sodium vessel. Also, because of doubts concerning the validity of extrapolating shock wave and corrosion effects from models, full scale tubes were used instead of 1/6 scale models in all subsequent tests.

During the remaining experiments, temperatures up to 1000°C were recorded near the tube burst, and rapid and severe corrosion of the failed tube and the surrounding bundle has been observed on several occasions. Significant deformation of the simulated tube bundle has also occurred, but this effect cannot be directly extrapolated to the PFR steam generator, whose

tube bundle will be structurally much stronger. Large quantities of reaction products were produced, and could not subsequently be drained from the reaction vessel.

In experiment no. 5 a secondary failure occurred in a pressurized tube adjacent to the fractured tube (Fig. 14), approximately 4.5 sec after the



FIG. 14. NOAH tube bundle after failure.

reaction was initiated. Examination of the tube bundle showed marked thinning of the failed tube and the tubes adjacent to it. It appears that the rupture in the failed tube was preceded by a considerable reduction in wall thickness, caused entirely by corrosion/erosion effects. The erosion thinning could have occurred at about 700°C , while the actual rupture of the failed tube apparently occurred between 850°C and 950°C . Similar failures occurred in experiments nos. 8 and 10.

During these experiments, the stresses in the reaction vessel were monitored with strain gauges, and showed that the pressure in the vessel did not exceed 20 kg/cm^2 during any experiment. The experiments therefore provide a reasonable basis for deciding the shell design pressure; they have also shown that an initial tube failure may lead to further tube failures, damage to the tube bundles, and formation of copious reaction products which could create problems when the steam generator is dismantled for repairs. Both the number of secondary tube failures and the quantity of corrosion products could be reduced if an initial tube failure were detected quickly, and devices to do this by detecting hydrogen both in the argon gas blanket and in the sodium are being developed at DERE.

7. Conclusions

The experience of sodium technology, gained by the operation of test rigs handling quantities of sodium up to the order of 10 m^3 at the maximum temperatures and fluid velocities envisaged, has established the feasibility of operating sodium circuits at PFR design conditions.

Studies of the PFR circuits have shown no engineering problems which would prevent realization of the design concept.

The additional information needed to establish the detailed design of components has been identified and the programme of engineering development to provide this information is well advanced.

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